

ACE uses causal mechanisms to choose experiments

Structural causal model (SCM)



Inputs

- Observed responses
- Known graph and legal actions
- Experiment budget

Outputs

- The next intervention
- Updated mechanism models
- Predictions for new actions

Each SCM equation describes one mechanism

$$X_i = f_i(Pa_i, U_i)$$

Parents Pa_i supply inputs. U_i represents outside influences.

$$X \rightarrow M \rightarrow Y$$

$$\begin{aligned} M &= 2.5X \\ Y &= 3M \end{aligned}$$

Worked example: known graph, unknown slopes, zero disturbances and dimensionless values.

An intervention replaces one mechanism

Ordinary system

$$\mathbf{X} \rightarrow \mathbf{M} \rightarrow \mathbf{Y}$$

Learn Y from measured M

Exclude the clamped M
label

Experiment: do(M = 1)

$$\mathbf{X} \quad \mathbf{M} = 1 \rightarrow \mathbf{Y}$$

One paid response can train several natural mechanisms. It is still one paid response.

Candidate experiments receive numerical scores

- 1 Simulate each legal intervention with the ensemble
- 2 Estimate how much it can reduce mechanism uncertainty
- 3 Select an action, then pay for its observed response

Score contribution = mean Cov^2 / (Variance + noise proxy)

Current PEV sums descendant contributions. Neural ensemble scores are estimates, not certified gains.

Exact inputs to the three-node demonstration

Mechanism	Member slopes	Mean prediction
M from X	1, 2, 3	$\hat{M} = 2X$
Y from M	2, 3, 4	$\hat{Y} = 3M$

Actions: clamp X or M to -1, 0 or +1

References: -1, 0, +1 Noise proxy: 0.05
Budget: one response Ties: first action in the list

Illustrative initialization. Fixed candidate grid. No exploration or candidate jitter in this example.

The score selects $\text{do}(X = -1)$

Intervention	M contribution	Y contribution	Total
$\text{do}(X = -1)$	0.413437	0.436264	0.849701
$\text{do}(X = 0)$	0.000000	0.000000	0.000000
$\text{do}(X = 1)$	0.413437	0.436264	0.849701
$\text{do}(M = -1)$	0.000000	0.413437	0.413437
$\text{do}(M = 0)$	0.000000	0.000000	0.000000
$\text{do}(M = 1)$	0.000000	0.413437	0.413437

Selected prediction: $M = -2$, $\hat{Y} = -6$

The two endpoint X actions tie. Choosing -1 follows the declared order, not observed performance.

The response updates eligible mechanisms

Observed row: $X = -1$, $M = -2.5$, $Y = -7.5$

Natural mechanism	Training input	Training label
M	Measured $X = -1$	$M = -2.5$
Y	Measured $M = -2.5$	$Y = -7.5$

After one illustrative gradient update

$M = -2.25$ **$\hat{Y} = -6.75$** **Budget left: 0**

This transparent linear update differs from production neural training. It proves no test-error gain.

A real ACE run records this first action

do(X7 = 3.857086181640625)

50

responses from the first intervention

Full campaign budget

1,600 intervention responses

400 observational responses

2,000 total responses

Seed 5000, first recorded step. The archive does not retain that step's raw rows or candidate scores.

The gains depend on the comparison

Question	Supported result	Important boundary
Mechanism prediction	Both fresh primary coverage contrasts pass	Known synthetic graphs Scoring ablation unresolved
Mechanism vs random	19 / 20 wins 64.85% lower mean MSE	Secondary comparison Primary coverage gate failed
Better final fitting?	11 / 12 histories improve Geometric error ratio 0.183	One worsens 2.015x Extra fitting, scored inits

Physical boundary: delivery beats Fourier in 0 of 11 conditions

Foundation models can propose reusable mechanisms

Pretrained numerical model or language proposer

SCM mechanisms with explicit intervention rules

Numerical checks and experiment selection

Test: fewer paid responses at matched target error

Proposed work. Compare strong data-only controls and deliberately wrong priors. Prior art already combines these ingredients.